

KUSHAL PATEL

Ann Arbor, MI 48103

📞 734-406-4636 ✉ patelku@umich.edu 👤 kushal5294.github.io

Education

University of Michigan

Ann Arbor, MI

MS in Computer Science, GPA: 3.8

May 2026

BS in Computer Science, GPA: 3.8

May 2025

- **Relevant Coursework:** Advanced Scalable Systems, Distributed Systems, Database Management Systems, Web Systems, Advanced Compilers, Computer Organization, Data Structures & Algorithms, Discrete Math, Theory of Computation, Game Theory, Robot Kinematics & Dynamics, Human-AI Interaction

Experience

Uber

July 2026 – Present

Software Engineer

Sunnyvale, CA

- Incoming Software Engineer.

Amazon

May 2025 – Aug 2025

Software Development Engineer Intern

Bellevue, WA

- Accelerated RGB-D camera calibration by 6x with a multithreaded, Linux C shared library that computes 4x4 transformation matrices for adjacent units via RANSAC plane detection & ICP edge-refined point cloud stitching.
- Designed a 10,000 TPS microservice that executes the calibration pipeline via AWS Lambda & DynamoDB.
- Extended library to automate re-calibration, adding CloudWatch alerts to sustain uptime across 185 Amazon centers.

University of Michigan, ITS

Jan 2024 – Apr 2024 & Aug 2024 – Dec 2024

Software Engineer Co-op

Ann Arbor, MI

- Improved image publishing pipeline by 66% for 50+ UM web developers by building 3 plugins that automatically generate alt text via generative AI from Azure OpenAI Service & OpenAI GPT-4o LLM.
- Released open source Google add-on, Drupal module, & WP plugin: <https://kushal5294.github.io/auto-alt.html>.

Next Play Games

May 2024 – Aug 2024

Software Engineer Intern

Los Angeles, CA

- Launched a fantasy baseball app by syncing MLB data for 2,087 players into a PostgreSQL database hosted on AWS.
- Built 8 REST API endpoints using Express.js and Redis to efficiently serve live stats, projections, and news to users.
- Ensured 100% app reliability by writing automated tests via Jest to validate daily box scores and player stats.

Projects

Sharded KVStore | Golang

- Achieved 100% linearizable consistency by running multi-instance Paxos across replicas to decide each log index.
- Enabled horizontal scaling by sharding keys into 16 groups & re-balancing server load via a centralized shardmaster.
- Ensured fault tolerance under replica crashes and network unreliability via RSM replay & duplicate RPC detection.

NFL Spread Predictor | Python, HTML, CSS

- Engineered a Pandas data frame of 304 metrics for all NFL games from past 7 years by web scraping with Selenium.
- Predicted score differential of future games by building a DNN via TensorFlow library & data frame.
- Accomplished 61% win rate and finished +37 units against DraftKings spread across 181 NFL games.
- Shared weekly picks by scripting a static site generator (Jinja2): <https://kushal5294.github.io/nfl-spread.html>.

Search Engine | Python, JavaScript, CSS

- Built inverted index of 2,936 Wikipedia pages by running tf-idf algorithm through MapReduce pipeline.
- Delivered top 10 page hits by creating Flask REST API to consult inverted index & PageRank score.
- Constructed frontend UI & deployed to AWS EC2: https://kushal5294.github.io/search_engine.html.

Skills

Languages: C++, Python, Go, Java, C, JavaScript, Typescript, HTML/CSS, SQL, Bash

Technologies: Hadoop, Docker, MySQL, MongoDB, NoSQL, Open3D

Concepts: Computer Vision, LLM, Operating Systems, OOP, Agile, Neural Networks, Unit Testing, Full Stack